

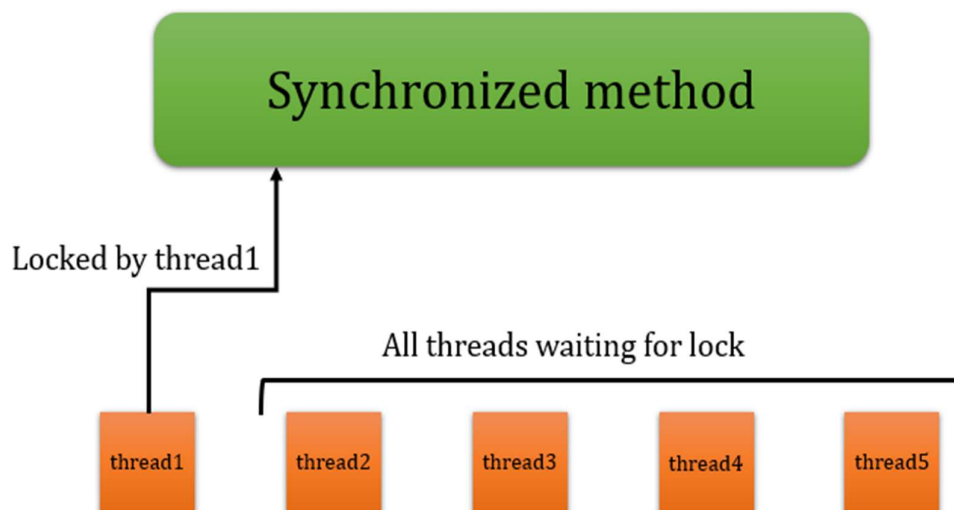
Synchronization in Java

✚ What is synchronization in java :-

- **Synchronization in Java** is a mechanism that ensures that **only one thread** can access a **shared resource (like a variable, method, or object)** at a time.
- It helps to **prevent data inconsistency and thread interference** when multiple threads try to modify shared data simultaneously.

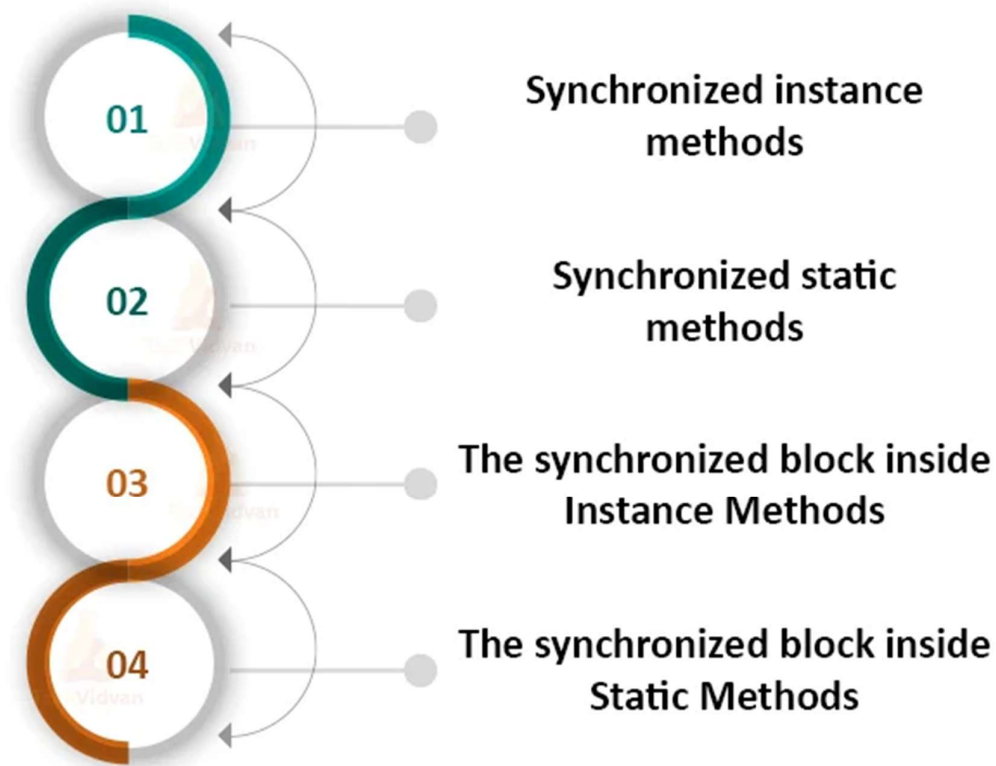
✚ Key Points in synchronization :-

- Synchronization is used in **multithreading** to control the access of multiple threads to shared resources.
- It ensures **thread safety** — meaning the output remains correct even when multiple threads run.
- You can synchronize using the **synchronized keyword** in Java.
- It can be applied to **methods** or **blocks of code**.



Methods in synchronization

Methods of Synchronization in Java



Example :-

```
class Example {  
    synchronized void display() {  
        System.out.println("Synchronized method example");  
    }  
}
```

Implementation of synchronization

Input :-

```
1 package Multi_Threading;
2
3 public class synchronization extends Thread{
4
5     public void run()
6     {
7         for(int i=1;i<=10;i++)
8         {
9             if(i%2==0)
10                System.out.println(Thread.currentThread().getName()+" :"+i);
11         }
12     }
13
14     public static void main(String[] args) throws InterruptedException {
15         synchronization t1 = new synchronization();
16         synchronization t2 = new synchronization();
17         synchronization t3 = new synchronization();
18
19         t1.start();
20         t1.join(); // Wait until thread 1 is completing the task
21         Thread.sleep(1000);
22         t2.start();
23         t2.join(); // Wait until thread is completing the task
24         Thread.sleep(1000);
25         t3.start();
26     }
27 }
```

Output :-

```
Thread-0 :2
Thread-0 :4
Thread-0 :6
Thread-0 :8
Thread-0 :10
Thread-1 :2
Thread-1 :4
Thread-1 :6
Thread-1 :8
Thread-1 :10
Thread-2 :2
Thread-2 :4
Thread-2 :6
Thread-2 :8
Thread-2 :10
```